CLASS 10 - MATHEMATICS

Sample Examination Paper

Time: 90 Minutes | Maximum Marks: 50

General Instructions:

1. All questions are compulsory.
2. Show necessary steps for calculations.
3. Use pi = 22/7 where required.
4. Calculators are not allowed.
5. Marks for each question are indicated in brackets.

SECTION A - Very Short Answer Questions — 5 x 2 = 10 marks

Q1. Find the HCF of 36 and 48 using prime factorization. **[2 marks]**

Q2. Solve: 3x - 7 = 11. **[2 marks]**

Q3. Find the distance between the points A(2, 3) and B(5, 7). **[2 marks]**

Q4. A die is thrown once. Find the probability of getting a prime number. **[2 marks]**

Q5. If the radius of a circle is 7 cm, find its circumference. **[2 marks]**

SECTION B - Short Answer Questions — 4 x 5 = 20 marks

Q6. The sum of two numbers is 27 and their difference is 5. Form a pair of linear equations and find the two numbers. **[5 marks]**

Q7. A ladder 10 m long reaches a window 8 m above the ground. Find the distance of the foot of the ladder from the wall. Also name the theorem used. **[5 marks]**

Q8. Find the mean of the following data: 6, 8, 10, 12, 14, 16, 18. **[5 marks]**

Q9. A cylinder has radius 3.5 cm and height 10 cm. Find its curved surface area. **[5 marks]**

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Sample Examination Paper - Continued

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SECTION C - Long Answer Questions — 20 marks

Q10(a). Solve the quadratic equation x^2 - 7x + 12 = 0 by factorization. **[5 marks]**

Q10(b). The first term of an arithmetic progression is 5 and the common difference is 3. Find the 20th term. **[5 marks]**

Q11. A rectangular garden is 30 m long and 20 m wide. A path of uniform width 2 m is constructed inside the garden along its boundary. Find: (i) the area of the path, and (ii) the cost of paving the path at Rs. 120 per square metre. **[10 marks]**

Space for rough work:

End of Question Paper